

The CVFEM Navier–Stokes operators

Discretisation of the incompressible equations on trilinear hexahedra, and the family of discrete operators a matrix-free solver evaluates

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Abstract

This note fixes the discretisation and enumerates the discrete operators built on it. The scheme is a colocated control-volume finite-element method on trilinear hexahedra: the unknowns live at the mesh nodes, the balance is written over the dual control volumes, and the fluxes are evaluated on sub-control surfaces cut out of each element. Pressure–velocity coupling is by a Rhie–Chow mass-flux correction; convection is upwinded through a switch that can be smoothed to keep the flux differentiable at stagnation.

The reason to write the operators down separately, rather than as one residual and its derivative, is that they are not one thing. A Newton–Krylov solve evaluates at least five of them per step, they carry different terms by design rather than by omission, and they differ in cost by a factor of fifty. Section 7 lists them with their domains of definition, and Section 8 with their measured throughput.

1 Continuous problem

On a bounded domain $\Omega \subset \mathbb{R}^3$ we solve the incompressible Navier–Stokes equations in conservative form for velocity $\mathbf{u}$ and pressure p , with constant density ρ and dynamic viscosity μ ,

$$\rho \frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) + \nabla p - \nabla \cdot \boldsymbol{\tau} = \rho \mathbf{f}, \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0, \quad (2)$$

with the Newtonian deviatoric stress

$$\boldsymbol{\tau} = \mu \left(\nabla \mathbf{u} + \nabla \mathbf{u}^\top \right), \quad \tau_{cd} = \mu (\partial_d u_c + \partial_c u_d). \quad (3)$$

The trace term $-\frac{2}{3}\mu(\nabla \cdot \mathbf{u})\mathbf{I}$ is absent: it vanishes for a divergence-free field and retaining it would make the discrete stress depend on how well (2) is satisfied at the current iterate rather than at the solution.

Integrating (1)–(2) over a control volume C with boundary ∂C and applying the divergence theorem gives the balance the discretisation reproduces:

$$\frac{d}{dt} \int_C \rho \mathbf{u} dV + \oint_{\partial C} [\rho \mathbf{u}(\mathbf{u} \cdot \mathbf{n}) + p \mathbf{n} - \boldsymbol{\tau} \mathbf{n}] dS = \int_C \rho \mathbf{f} dV, \quad (4)$$

$$\oint_{\partial C} \rho \mathbf{u} \cdot \mathbf{n} dS = 0. \quad (5)$$

Every term of every operator below is a quadrature of one of these two, so the discrete system is conservative by construction: a flux leaving one control volume enters its neighbour with the opposite sign and the same value.

2 The dual mesh and its geometry

2.1 Control volumes and sub-control surfaces

The primal mesh is a partition of Ω into trilinear hexahedra. The unknowns $(\mathbf{u}_i, p_i)$ are colocated at the N mesh nodes, and the control volume C_i is the median-dual cell of node i : within each element containing i , the sub-volume bounded by the planes through the element centroid, the face centroids and the edge midpoints. Each hexahedron therefore contributes one of its eight sub-volumes to each of its eight nodes, and

$$V_i = \sum_{e \ni i} \frac{1}{8} |\det \mathbf{J}_e|, \quad (6)$$

where $\mathbf{J}_e$ is the element Jacobian defined below. The V_i partition Ω exactly: $\sum_i V_i = |\Omega|$.

The interface between C_i and C_j inside one element is a *sub-control surface* (SCS). In a hexahedron the pairs (i, j) that share such an interface are exactly the twelve edges, so each element carries $N_{\text{scs}} = 12$ interior sub-control surfaces, grouped four to each of the three parametric directions. A sub-control surface lying on $\partial\Omega$ is instead a *boundary sub-face*; each of the six element faces carries four of them, one per node of that face, and they close the control volumes that touch the domain boundary.

2.2 Isoparametric map and area vectors

Let $\mathbf{x}_a$, $a = 0, \dots, 7$, be the nodes of an element and $N_a(\boldsymbol{\xi})$ the trilinear shape functions on the reference cube $\hat{\Omega} = [0, 1]^3$. The geometric map and its Jacobian are

$$\mathbf{x}(\boldsymbol{\xi}) = \sum_a N_a(\boldsymbol{\xi}) \mathbf{x}_a, \quad \mathbf{J}(\boldsymbol{\xi}) = \frac{\partial \mathbf{x}}{\partial \boldsymbol{\xi}}, \quad \mathbf{A}(\boldsymbol{\xi}) = \text{adj } \mathbf{J} = \det(\mathbf{J}) \mathbf{J}^{-1}. \quad (7)$$

Gradients push forward as $\nabla_{\mathbf{x}} \phi = \mathbf{J}^{-\text{T}} \nabla_{\boldsymbol{\xi}} \phi = \mathbf{A}^{\text{T}} \nabla_{\boldsymbol{\xi}} \phi / \det \mathbf{J}$.

The sub-control surface associated with parametric direction d is, in the reference element, a quarter of the unit face normal to ξ_d . By Nanson's formula its oriented area in physical space is

$$\mathbf{a}^{(d)} = \int_{\hat{S}} \det(\mathbf{J}) \mathbf{J}^{-\text{T}} \mathbf{e}_d d\hat{S} = \frac{1}{4} \mathbf{A}_{d,:} = \frac{1}{4} \det(\mathbf{J}) \nabla \xi_d, \quad (8)$$

the d -th row of the adjugate, scaled by the reference sub-face area $\frac{1}{4}$. This single identity supplies every area vector in the scheme, interior and boundary alike; a boundary sub-face on the element face normal to ξ_d carries $\pm \mathbf{a}^{(d)}$ with the sign chosen outward.

2.3 Two geometric variants

Equation (8) is evaluated in two ways, and the choice is a modelling decision rather than an optimisation.

Isoparametric. $\mathbf{J}$, $\mathbf{A}$ and $\det \mathbf{J}$ are evaluated at each sub-control surface centroid $\boldsymbol{\xi}_s$, and the velocity gradient with them. Each of the twelve surfaces then carries its own area vector and its own gradient. This is exact for the trilinear geometry.

Affine. The geometry is evaluated once, at the element centre $\boldsymbol{\xi} = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, and reused for all twelve surfaces. The four sub-control surfaces of a given direction then share one area vector $\mathbf{a}^{(d)}$, and the velocity gradient is a single element constant

$$\nabla \mathbf{u}|_e = \frac{1}{\det \mathbf{J}} \mathbf{A}^{\text{T}} [\delta_1 \mathbf{u} \ \delta_2 \mathbf{u} \ \delta_3 \mathbf{u}], \quad \delta_d \mathbf{u} = \frac{1}{4} \sum_{a \in F_d^+} \mathbf{u}_a - \frac{1}{4} \sum_{a \in F_d^-} \mathbf{u}_a, \quad (9)$$

with $F_d^\pm$ the four nodes of the two element faces normal to ξ_d . For a parallelepiped the two variants coincide identically; on a distorted hexahedron the affine variant is a one-point rule for a bilinear surface and is second-order accurate rather than exact. Both are carried, because the affine form collapses twelve geometric evaluations into one and the resulting operator is between three and five times cheaper.

3 The interior flux

Let s be a sub-control surface of an element, separating nodes i and j , with area vector $\mathbf{a}$ oriented from i to j , and write

$$\bar{\mathbf{u}} = \frac{1}{2}(\mathbf{u}_i + \mathbf{u}_j), \quad \bar{p} = \frac{1}{2}(p_i + p_j), \quad \mathbf{d} = \mathbf{x}_j - \mathbf{x}_i. \quad (10)$$

3.1 Mass flux

The mass flux through s is the linearly interpolated normal flux plus the Rhie–Chow correction of §4,

$$\dot{m}_s = \rho \bar{\mathbf{u}} \cdot \mathbf{a} + \dot{m}_s^{\text{RC}}. \quad (11)$$

3.2 Convection, pressure and viscous traction

The convective momentum flux is upwinded on the sign of $\dot{m}_s$,

$$\dot{m}_s^+ = \frac{1}{2}(\dot{m}_s + |\dot{m}_s|_\varepsilon), \quad \dot{m}_s^- = \frac{1}{2}(\dot{m}_s - |\dot{m}_s|_\varepsilon), \quad (12)$$

so that $\dot{m}_s^+ \mathbf{u}_i + \dot{m}_s^- \mathbf{u}_j$ selects the upwind node exactly when $|\cdot|_\varepsilon = |\cdot|$. The pressure is interpolated centrally and the viscous traction is taken from the element gradient of §2.3. The total momentum and mass fluxes through s are

$$\begin{aligned} \mathbf{F}_s &= \dot{m}_s^+ \mathbf{u}_i + \dot{m}_s^- \mathbf{u}_j + \bar{p} \mathbf{a} - \boldsymbol{\tau} \mathbf{a}, \\ \boldsymbol{\tau} \mathbf{a} &= \mu \left[\left(\nabla \mathbf{u} + \nabla \mathbf{u}^\top \right) \mathbf{a} \right]. \end{aligned} \quad (13)$$

Each surface contributes to both of its nodes with opposite signs,

$$\mathbf{R}_i += \mathbf{F}_s, \quad \mathbf{R}_j -= \mathbf{F}_s, \quad R_i^c += \dot{m}_s, \quad R_j^c -= \dot{m}_s, \quad (14)$$

which is what makes the scheme discretely conservative: summing R^c over all nodes leaves only boundary sub-faces.

3.3 The smoothed upwind switch

The hard switch takes $|\dot{m}|_\varepsilon = |\dot{m}|$, which has a corner at $\dot{m} = 0$. Where the flow stagnates — inside a recirculation, or across a partly reversed outlet — sub-control surfaces sit arbitrarily close to that corner while the two upwinded states still differ by $O(1)$, so the flux is not differentiable near the iterate and Newton loses its rate. Following Venkatakrishnan’s requirement that a switch be inactive where it is not needed and reach that state smoothly, the absolute value is replaced by

$$|\dot{m}|_\varepsilon = \begin{cases} |\dot{m}|, & |\dot{m}| \geq \varepsilon, \\ \frac{\dot{m}^2 (2\varepsilon - |\dot{m}|)}{\varepsilon^2}, & |\dot{m}| < \varepsilon, \end{cases} \quad (15)$$

which matches $|\dot{m}|$ in value and slope at $|\dot{m}| = \varepsilon$, vanishes with zero slope at the origin, and never exceeds $|\dot{m}|$. Surfaces carrying real flux are untouched; surfaces carrying none lose their upwinding and blend to central differencing. The direction matters: an entropy-fix style band that saturates $|\dot{m}|$ at a floor $\varepsilon/2$ instead adds dissipation $\frac{\varepsilon}{4}(\mathbf{u}_i - \mathbf{u}_j)$ on every zero-flux surface, which in a unidirectional flow is most of the mesh. Taking $\varepsilon = 0$ recovers the hard switch exactly.

3.4 Second order by deferred correction

The upwinded state in (13) is the donor node’s value, which is first order. The scheme reaches second order by a deferred correction: the donor value is extrapolated to the sub-control surface along the reconstructed nodal gradient,

$$\phi_{\text{up}} = \phi_C + \Psi \nabla \phi_C \cdot \mathbf{d}_{C \rightarrow s}, \quad (16)$$

with $\mathbf{d}_{C \rightarrow s}$ the offset from the donor node to the surface centroid and Ψ a limiter. The gradient is the same nodal reconstruction the Rhie–Chow term uses (Section 4.1), so the correction adds no sweep of its own; it is lagged one Newton step, which is what “deferred” means here.

Four choices of Ψ are carried, because the limiter is the part of the reconstruction with data-dependent selects in it and each behaves differently:

- $\Psi = 1$, unlimited, which is second order and not bounded;
- a bounded-face clip, which pins ϕ_{up} into the interval spanned by the two nodes of the surface;
- Venkatakrishnan’s limiter, with his dimensionless K setting $\varepsilon^2 = u_{\text{ref}}^2 (K/L)^3 h^3$ so that the limiter switches itself off where the field is smooth;
- Darwish and Moukalled’s, which rebuilds the one-dimensional TVD stencil by placing a virtual upwind node, so the classical limiter family applies with no neighbour min/max.

Two properties matter for the Jacobian rather than for the solution. The limited increment is continuous but not differentiable: a clip is non-smooth where it engages, and both remaining limiters branch on a sign. The Jacobian action differentiates (16) exactly, which means differentiating through the reconstruction — the same second pass the exact Rhie–Chow term pays for — and means choosing a subgradient where the limiter has none. Those choices have to be made against a rounding of the data and not against a quantity that vanishes with the one being tested, or two implementations of the same operator disagree by far more than round-off: the measured consequence of getting it wrong was 10^{-4} between two mesh layouts, on a limiter whose value agreed to the last bit. And $K = 0$, which is a legitimate limiter for the value, is the case in which Venkatakrishnan’s ε^2 stops bounding that choice.

4 Pressure–velocity coupling

On a colocated arrangement the mass flux $\rho \bar{\mathbf{u}} \cdot \mathbf{a}$ does not see the odd pressure mode, and (5) admits a checkerboard. The Rhie–Chow correction restores the coupling by adding to the interpolated flux a term proportional to the difference between the *compact* pressure difference across the surface and the *reconstructed* one,

$$\dot{m}_s^{\text{RC}} = -c_s [(p_j - p_i) - \tfrac{1}{2} (\nabla p_i + \nabla p_j) \cdot \mathbf{d}], \quad (17)$$

with the surface coefficient

$$c_s = \rho D_s \frac{|\mathbf{a}|^2}{\mathbf{a} \cdot \mathbf{d}}, \quad D_s = \frac{\alpha |\mathbf{d}|^2}{2\mu}, \quad \text{hence} \quad c_s = \frac{\alpha \rho |\mathbf{d}|^2 |\mathbf{a}|^2}{2\mu (\mathbf{a} \cdot \mathbf{d})}. \quad (18)$$

Here α is a dimensionless scale, $\alpha = 1$ by default and $\alpha = 0$ recovering the uncorrected flux. The coefficient is set to zero on a degenerate surface, $|\mathbf{a} \cdot \mathbf{d}|$ below a relative floor, so that a collapsed element contributes nothing rather than a division by a vanishing quantity.

Two properties of (17) are worth stating because they determine where it may be used. First, the bracket annihilates any pressure field whose reconstructed gradient is exact, in particular any globally linear p : the correction is a genuine stabilisation and not a consistency error. Second, c_s depends only on the geometry and on ρ, μ, α , so it is constant over a Newton solve and may be precomputed once per surface.

4.1 The reconstructed nodal gradient

The gradients in (17) are the volume-weighted average of the element gradients meeting at the node,

$$\nabla p_i = \frac{\sum_{e \ni i} |\det J_e| \nabla p|_e}{\sum_{e \ni i} |\det J_e|}, \quad (19)$$

where $\nabla p|_e$ is the element-constant gradient (9) applied to p , or its isoparametric counterpart at the element centre. This is a linear operator $\mathcal{G} : \mathbb{R}^N \rightarrow \mathbb{R}^{3N}$ acting on the nodal pressures, and its stencil reaches every node of every element containing i — a wider set than the nearest-neighbour graph of the mesh. That single fact separates two of the operators in §7.

5 Boundary closure

A control volume touching $\partial\Omega$ is not closed by interior sub-control surfaces alone; its boundary sub-faces must carry the remaining flux, or (5) is not a balance at all. For a boundary sub-face of node i with outward area vector $\mathbf{a}$ the flux is $\dot{m} = \rho \mathbf{u}_i \cdot \mathbf{a}$ and three closures are used.

Closed / prescribed velocity. The full flux with the local state,

$$\mathbf{R}_i += \dot{m} \mathbf{u}_i + p_i \mathbf{a} - \boldsymbol{\tau} \mathbf{a}, \quad R_i^c += \dot{m}. \quad (20)$$

On a no-slip wall $\mathbf{u}_i = 0$ makes the convective and mass terms vanish identically and leaves the pressure and viscous traction, which is the correct wall contribution.

Prescribed pressure. As (20) with $\bar{p}$ in place of p_i . The viscous traction is still evaluated from the interior state, so this prescribes the pressure and not the whole normal traction.

Natural (do-nothing) outflow. The condition $(p\mathbf{l} - \boldsymbol{\tau})\mathbf{n} = \mathbf{t}$ is imposed by *omitting* the pressure and viscous terms rather than evaluating them,

$$\mathbf{R}_i += \max(\dot{m}, 0) \mathbf{u}_i + \mathbf{t} |\mathbf{a}|, \quad R_i^c += \dot{m}. \quad (21)$$

Two details are load-bearing. Retaining $p_i \mathbf{a}$ here would leave the constant-pressure mode in the null space — a uniform pressure shift integrates to zero over every closed control volume — so the gauge would remain undetermined and require a pin. And the convective term is clipped to $\max(\dot{m}, 0)$ because this closure has no upwind switch: on a sub-face with $\dot{m} < 0$ the unguarded form convects the exterior state inward, which is the classical backflow instability of a recirculating outlet. The continuity row keeps the true, unclipped $\dot{m}$, since clipping it would destroy global mass conservation.

6 Time integration and body force

The transient term is the only place where the control-volume structure makes the mass matrix trivial. In (4) the volume integral of $\rho \mathbf{u}$ over C_i is $\rho V_i \mathbf{u}_i$ up to the same order as the rest of the scheme, so the mass matrix *is* the control volume: diagonal, already available from (6), and requiring no assembly. A consistent finite-element mass matrix would be a different discretisation, not a better one.

With backward differentiation of order k and step Δt ,

$$\mathbf{R}_i += \frac{\rho V_i}{\Delta t} (a_0 \mathbf{u}_i^{n+1} + a_1 \mathbf{u}_i^n + a_2 \mathbf{u}_i^{n-1}), \quad \begin{array}{ll} \text{BDF1:} & a_0 = 1, \quad a_1 = -1, \quad a_2 = 0, \\ \text{BDF2:} & a_0 = \frac{3}{2}, \quad a_1 = -2, \quad a_2 = \frac{1}{2}, \end{array} \quad (22)$$

with $\sum_k a_k = 0$, so a constant history contributes exactly nothing. The term touches the three velocity components and never the continuity row: the incompressibility constraint has no time derivative, and giving it one would be a different system. Its contribution to the Jacobian is therefore the diagonal

$$\frac{\partial}{\partial \mathbf{u}_i} \left[\frac{\rho V_i a_0}{\Delta t} \mathbf{u}_i \right] = \frac{\rho V_i a_0}{\Delta t} \mathbf{I}_{3 \times 3}, \quad (23)$$

independent of the history. A body force enters the same way, as $\mathbf{R}_i = \rho \mathbf{f}(\mathbf{x}_i) V_i$, and for the same reason: neither term involves a sub-control-surface flux, so both are node-local post-passes over the mesh.

7 The discrete operators

Collect the unknowns as $w = (\mathbf{u}_1, p_1, \dots, \mathbf{u}_N, p_N) \in \mathbb{R}^{4N}$. Everything above defines one nonlinear map; what a solver actually evaluates is the family below. They are listed with what each carries, because they do not all carry the same terms and the differences are deliberate.

7.1 $\mathcal{F}$ — the residual

$$\mathcal{F}(w)_i = \underbrace{\sum_{s \in \partial C_i} \pm \mathbf{F}_s}_{\text{interior sub-control surfaces}} + \underbrace{\sum_b \mathbf{F}_b}_{\text{boundary closure}} + \underbrace{\frac{\rho V_i}{\Delta t} \sum_k a_k \mathbf{u}_i^{n+1-k}}_{\text{transient}} - \underbrace{\rho \mathbf{f}_i V_i}_{\text{body force}}, \quad (24)$$

with the continuity component assembled from $\pm \dot{m}_s$ and $\dot{m}_b$ in the same sweep. Evaluating it requires the nodal gradient (19) of the current pressure, which is a function of the Newton iterate and may therefore be computed once per Newton step rather than once per residual.

7.2 $\mathcal{J}(w)v$ — the exact Jacobian action

The Fréchet derivative applied to a direction $v = (\mathbf{v}, q)$. Every term of (24) is differentiated, including the reconstruction (19): since $\mathcal{G}$ is linear, its derivative in the direction q is $\mathcal{G}q$, and the Rhie–Chow perturbation is

$$\delta \dot{m}_s^{\text{RC}} = -c_s [(q_j - q_i) - \tfrac{1}{2} (\nabla q_i + \nabla q_j) \cdot \mathbf{d}], \quad \nabla q = \mathcal{G}q. \quad (25)$$

The upwind switch is differentiated through (15), whose derivative is

$$\frac{d|\dot{m}|_\varepsilon}{d\dot{m}} = \begin{cases} \text{sgn}(\dot{m}), & |\dot{m}| \geq \varepsilon, \\ \text{sgn}(\dot{m}) \frac{4\varepsilon|\dot{m}| - 3\dot{m}^2}{\varepsilon^2}, & |\dot{m}| < \varepsilon. \end{cases} \quad (26)$$

This operator is the one a Krylov method applies, and (25) is what makes it expensive: $\mathcal{G}q$ is a full sweep over the elements, the direction changes at every iteration, and so — unlike $\mathcal{G}p$ — it cannot be hoisted out of the solve.

7.3 $J(w)$ — the assembled Jacobian, with frozen reconstruction

The same derivative with the second term of (25) dropped,

$$\delta \dot{m}_s^{\text{RC}} = -c_s (q_j - q_i), \quad (27)$$

that is, the reconstruction is held frozen at the current iterate. This is not an oversight and not an approximation made for speed: by §4.1 the operator $\mathcal{G}$ couples node i to every node of every element containing i , so retaining it would widen the sparsity pattern beyond the mesh’s nearest-neighbour graph. The assembled matrix exists to build preconditioners, and a preconditioner on

a wider graph than the operator it preconditions is a poor trade. With (27) the pattern is exactly the node-to-node graph and the matrix has the 4×4 block structure

$$J = \begin{pmatrix} F & B^\top \\ B & C \end{pmatrix}, \quad (28)$$

with F the momentum block (convection, viscosity and, when unsteady, the mass term), B the discrete divergence and $B^\top$ the discrete gradient. The pressure block carries a plus sign here because the continuity residual is written as $+\oint \rho \mathbf{u} \cdot \mathbf{n}$; the $-C$ familiar from the stabilised-Stokes literature is the same operator with that row negated.

C is the Rhie–Chow operator, and differentiating (27) through the scatter (14) identifies it exactly: $C_{ij} = -c_s$ for each sub-control surface joining i and j , and $C_{ii} = \sum_{s \ni i} c_s$. It is the graph Laplacian of the mesh with weights $c_s > 0$ — symmetric, positive semidefinite, with the constant vector in its kernel, which is the discrete statement that the pressure is determined only up to a gauge.

Without Rhie–Chow the fourth row and column of J do not merely lose their diagonal: they are structurally empty at every node whose control volume is closed. Summing the oriented areas over a closed surface gives $\sum_{s \in \partial C_i} \sigma_s \mathbf{a}_s = 0$, and both B_{ii} and $B_{ii}^\top$ are $\frac{\ell}{2}$ and $\frac{1}{2}$ times exactly that sum. So $C = 0$ leaves a genuine saddle point whose 4×4 diagonal blocks are singular with a one-dimensional kernel — which is why no point- or block-wise relaxation can be built on them, and why the boundary closure of §5 is a prerequisite for the preconditioner and not only for conservation.

7.4 D — the block diagonal

The N diagonal 4×4 blocks of (28),

$$D_i = \frac{\partial \mathcal{F}_i}{\partial w_i} = \begin{pmatrix} F_{ii} & B_{ii}^\top \\ B_{ii} & C_{ii} \end{pmatrix} \in \mathbb{R}^{4 \times 4}, \quad (29)$$

carrying the same terms as J , including the boundary closure and the transient diagonal (23). It is the object a block-Jacobi smoother inverts, and by the closure identity of §7.3 its usefulness rests entirely on $C_{ii} \neq 0$: on a closed control volume the other three entries of its fourth row and column vanish, so without the Rhie–Chow term D_i is singular for every i and a smoother built on it is undefined rather than merely slow.

7.5 Jv — the sparse matrix–vector product

Given J assembled, the action can also be evaluated as a block sparse matrix–vector product. Mathematically this is §7.3 applied to v — it therefore carries the frozen reconstruction (27) and differs from §7.2 by exactly the term $\frac{1}{2}c_s(\nabla q_i + \nabla q_j) \cdot \mathbf{d}$. Its cost is set by the sparsity pattern alone and is independent of which physical terms the entries carry.

7.6 S — the pressure Schur complement diagonal

For a SIMPLE-type preconditioner the exact Schur complement $C - BF^{-1}B^\top$ is replaced by its diagonal, with F approximated by its own velocity diagonal,

$$S_i = \text{diag}\left(C - B \text{diag}(F)^{-1} B^\top\right)_i, \quad (30)$$

read directly from the assembled blocks: C_{ii} is the pressure–pressure entry of block (i, i) , and the correction sums over the row of J . Unlike the classical μM_p^{-1} approximation this makes no assumption about which term dominates — it reads both the Rhie–Chow operator and the velocity coupling out of the matrix that was assembled.

7.7 Splitting the momentum block

The viscous contribution to (13) is $-\tau \mathbf{a}$ with τ linear in $\mathbf{u}$ and $\mathbf{a}$ purely geometric, so that part of F depends on the mesh and μ alone and is constant across a Newton solve. Writing $F = F_{\text{visc}} + F_{\text{conv}}(\mathbf{u})$ allows the constant half to be formed once and the varying half added at each iteration; the sum reproduces F exactly, since the two are the two halves of one expression rather than two approximations. The Rhie–Chow blocks belong entirely to the varying half.

8 What the terms cost

The operators above differ in cost by more than an order of magnitude, and the differences do not follow the number of terms. The table gives measured throughput in millions of degrees of freedom per second on one 72-core Grace socket at 4 121 204 degrees of freedom, on the packed mesh layout, with the bare element sweep — no Rhie–Chow, no boundary closure, no transient — as the reference. Each figure is the median of five runs in one allocation; the rows re-measured at eleven runs carried a 1–2% spread.

Operator	MDOF/s	vs. bare sweep
Residual $\mathcal{F}$, interior flux only	2839	1.00
+ Rhie–Chow, reconstruction reused	1834	1.55
+ boundary closure	1640	1.73
+ transient	1519	1.87
+ reconstruction rebuilt every evaluation	1263	2.25
Jacobian action $\mathcal{J}v$, interior flux only	2055	1.00
+ exact Rhie–Chow (25)	944	2.18
+ boundary closure and transient	839	2.45
the same with the frozen Rhie–Chow form	1179	1.74
Matrix–vector product Jv	440	4.67
Assembly of J	39–103	28–73
Block diagonal D	113	—

The assembly row spans the four scatter layouts — 39 on the flat atomic sweep, 63 packed, 88 with the streaming flush, 103 with the pack colouring — and the layout matters more here than any term does. For comparison, the standard atomic layout runs the bare residual at 1135 against the packed layout’s 2839, a factor of 2.5, and every row of the table keeps that factor to within a few percent.

The frozen-Rhie–Chow row is there because the assembled matrix carries that form and not the exact one: it cannot carry the exact term without widening its one-ring sparsity. It is therefore the matrix-free operator to compare against Jv , and the only row in this table that is not part of the cumulative sequence above it.

8.1 The higher-order convective flux

Every row above runs the first-order upwind flux. The deferred correction of Section 3.4 is what the scheme runs when accuracy matters, and it is the one part of the operator family the table said nothing about. Measured in one allocation at the same size, medians of five, with the first-order Rhie–Chow operator as the reference because that is what each correction is added to — 1821 and 950 here, which agree with the corresponding rows above to within 0.7%.

Operator	MDOF/s	vs. first order	flops/element
Residual, first order + Rhie–Chow	1821	1.00	754
+ correction, unlimited	1108	1.64	1558
+ correction, bounded-face clip	856	2.13	1702
+ correction, Venkatakrishnan	748	2.44	2638
+ correction, Darwish–Moukalled	877	2.08	2278
+ Venkatakrishnan, boundary and transient	683	2.67	2638
Jacobian action, first order + exact Rhie–Chow	950	1.00	850
+ correction, unlimited	449	2.12	2422
+ correction, Venkatakrishnan	328	2.89	5878
+ Venkatakrishnan, boundary and transient	316	3.00	5878

The flop column is the driver’s model, counted against the kernel line for line. It includes the reconstruction of the twenty-four node-to-surface increments once per element, which on the affine element is a product with the three edge columns of the Jacobian rather than a centroid built from eight coordinates, so no coordinate is read by the correction at all. The correction is a lane-blocked arm of the same face kernel as the flux it is added to and shares its gathers and scatters, which is why its throughput ratio of 2.44 for Venkatakrishnan sits below the flop ratio of 3.5: the arithmetic grows faster than the time.

The packed layout’s advantage shrinks here, from 1.95 over the standard layout for the first-order residual to 1.37 for the Venkatakrishnan one. The correction adds arithmetic that both layouts perform alike; the layout’s gain is in the scatter.

The derivative is the expensive half, and by more than the residual’s factor: the correction costs the Jacobian action 2.89 against the residual’s 2.44 on the same limiter, on top of an operator that was already 2.18 times the bare sweep, and its flop count rises from 850 to 5878 — a factor of 6.9, against the residual’s 3.5. The limiter has to be differentiated through the nodal gradient reconstruction, and that reconstruction is what the exact Rhie–Chow term already pays for once.

Three readings of this table bear on the design rather than on the implementation.

The exact Rhie–Chow term costs the Jacobian action a factor of 2.2 on its own, against 1.55 for the same term in the residual. The asymmetry is (25): the residual reconstructs the gradient of a field that is fixed over the Newton step, the action reconstructs the gradient of a direction that changes at every Krylov iteration, and only the former can be amortised. Where the reconstruction cannot be reused — the last residual row — the residual pays the same factor.

The matrix–vector product is slower than the matrix-free action carrying the same physics, by 2.7, before counting the assembly that produced the matrix — and “the same physics” has to mean the frozen Rhie–Chow form, since that is what the assembled matrix encodes. The exact action, which the solver actually applies, is 1.9 times the product. The reason is traffic rather than arithmetic: the assembled blocks are roughly 840 bytes per degree of freedom and must be streamed in full at every application, whereas the matrix-free operator reads the state and the mesh and recomputes the rest. This is also why the product’s cost is flat in the physics while the matrix-free operator’s is not. The assembled matrix earns its place as the source of D and S , not as a way to apply J .

Adding physics narrows the gap between matrix-free and assembled from 4.7 to 2.7 without closing it. Any future term heavy enough to close it would also, by the same argument, be heavy enough to make assembly worth revisiting — which is the quantitative form of the usual matrix-free trade-off, and the reason the terms are enumerated separately here rather than folded into a single residual.